Journal of Analytical and Applied Pyrolysis 166 (2022) 105611

Contents lists available at ScienceDirect

Journal of Analytical and Applied Pyrolysis

journal homepage: www.elsevier.com/locate/jaap

Fractionation by flash chromatography and molecular characterization of bio-oil by ultra-high-resolution mass spectrometry and NMR spectroscopy

Jasmine Hertzog [a,b,c,*,1], Carole Garnier [d], Charlotte Mase [a,b,d,2], Sandra Mariette [d], Olivier Serve [d,3], Marie Hubert-Roux [a,b,4], Carlos Afonso [a,b,5], Pierre Giusti [a,b,d,6], Caroline Barrère-Mangote [b,d,7]

[a] Normandie Université, COBRA, UMR 6014 et FR 3038, Université de Rouen, INSA de Rouen-Normandie, CNRS, IRCOF, 76821 Mont-Saint-Aignan Cedex, France
[b] International Joint Laboratory – iC2MC: Complex Matrices Molecular Characterization, TRTG, BP 27, 76700 Harfleur, France
[c] Université de Lorraine, LCP-A2MC, 57000 Metz, France
[d] TotalEnergies OneTech, TotalEnergies Research & Technology Gonfreville, BP 27, 76700 Harfleur, France

ARTICLE INFO

Keywords:
Flash chromatography
Fractionation
Bio-oil
FT-ICR MS
NMR

ABSTRACT

Bio-oils obtained from the pyrolysis of lignocellulosic biomass are highly complex matrices with tens of thousands of organic compounds covering a wide polarity and mass range. Their upgrading into greener fuel or value-added chemicals requires an in-depth knowledge of their molecular composition prior and after valorization treatment, which can be obtained by different analytical techniques. Here, flash chromatography was employed as an efficient way to fractionate a bio-oil vacuum residue into four distinct fractions. This process was repeatable at both gravimetric and chromatographic levels and provided information about the weight contribution of each fraction to bio-oil composition. Achieved fractions were then characterized by ultra-high-resolution Fourier transform ion cyclotron resonance mass spectrometry (FT-ICR MS) and nuclear magnetic resonance spectroscopy. These two techniques ensured the good repeatability of bio-oil fractionation at the molecular level and also highlighted molecular specificities of each fraction. Thus, flash chromatography provided quantitative information that was complementary to FT-ICR MS results. It also simplifies the bio-oil compositional analysis, making it possible to detect more compounds. Eventually, this fractionation process can also be used to target fraction or compounds of interest for bio-oil upgrading.

1. Introduction

Thermochemical conversion, such as pyrolysis and liquefaction, of lignocellulosic biomass notably yields a liquid product called bio-oil. This latter material is a promising source of renewable fuels and value-added chemicals [1–3]. However, bio-oils are highly complex matrices, with tens of thousands of organic compounds and a high oxygen content, which are responsible for corrosion and storage issues and prevent bio-oil from being directly used as transportation fuel [4–7]. Therefore, upgrading processes must be applied, such as hydrodeoxygenation, to obtain a so-called second-generation or advanced biofuels [8,9]. However, both conversion and upgrading steps still need to be optimized for biofuel production. To monitor and improve these processes, the most detailed composition description of the raw and upgraded bio-oils is required. Several methods exist for bio-oil characterization that involve chromatography, spectroscopy, and mass spectrometry [10–12]. Recently, our research group reported that sole direct infusion high-resolution mass spectrometry (HRMS) enables the

* Correspondence to: Université de Lorraine, LCP-A2MC, 1 Boulevard Arago, 57078 Metz, France.
E-mail address: jasmine.hertzog@univ-lorraine.fr (J. Hertzog).
[1] 0000-0002-7946-1904.
[2] 0000-0001-8820-2595.
[3] 0000-0002-0199-8453.
[4] 0000-0002-0199-8453.
[5] 0000-0002-2406-5664.
[6] 0000-0002-9569-3158.
[7] 0000-0001-6774-2509.

https://doi.org/10.1016/j.jaap.2022.105611
Received 22 April 2022; Received in revised form 8 July 2022; Accepted 9 July 2022
Available online 14 July 2022
0165-2370/© 2022 Elsevier B.V. All rights reserved.

detection and the assignment of tens of thousands of bio-oil components, but only at the isobaric level [13]. In fact, isomeric information was showed to be relevant in the follow-up and understanding of the upgrading treatment of energetic resources such as crude oil [14]. A way to access to that level of information is to combine chromatography or ion-mobility spectrometry with HRMS [15–18]. Indeed, offline or online separation step prior HRMS provides better insight into bio-oil composition with isomer differentiation, but also enables identification of hitherto unknown species [16,19]. Upstream separation actually reduces bio-oil complexity and ion suppression phenomenon caused by direct infusion mode.

Besides biofuel valorization, bio-oil can be upgraded based on its compounds or fractions of interest, which therefore requires a separation process [2,3,20]. For single compounds, their concentration in the bio-oil must be high enough to be separated, this is the case of acetic acid, acetol, hydroxyacetaldehyde, and levoglucosan. These species can be intermediates in chemical synthesis, and also used in the food and textile industries [3,21]. Regarding bio-oil fractions, other applications can be considered. Thus, pyrolytic lignin and phenol compounds can be used as a source of antioxidants and resins while water-soluble fraction contains components usable as food additives and in hydrogen production [3,22–24].

Hence, bio-oil separation is a key step in bio-oil analysis, refining, and upgrading [25,26]. Different separation technologies with their characteristics were reported [2,3,22]. For instance, liquid-liquid extraction, distillation, and membrane separation can be applied. Solid-liquid chromatography (SLC) and, more specifically, adsorption chromatography is an interesting method compared to liquid-liquid extraction as there is no emulsion formation between bio-oil and solvent and it can be automated for better repeatability and reproducibility. Principles, advantages, and challenges of the adsorption chromatography, as well as a review of the existing studies related to bio-oil, were reported by Chan et al.[26]. Different adsorbents can be used depending on the upgrading purposes. For instance, Stanford et al. studied two polymeric resin adsorbents, SP207 and XAD4, for the separation of phenolic species over levoglucosan [27]. But most of the studies were performed on silica column. Thus, Pütün et al. fractionated the bio-oil into pentane-soluble fraction by SLC. Silica gel column was successively eluted with pentane, toluene, ether, and methanol to respectively recover aliphatic, aromatic, ester, and polar fractions [28]. In another study, the same process was applied, without the ester fraction, as well as Ozcimen et al. [29,30]. Other mobile phases can be used to elute bio-oil components such as hexane/ethyl acetate followed by chloroform/methanol [31]. Ware et al. performed bio-oil silica gel fractionation using cyclohexane, cyclohexane/dichloromethane, dichloromethane/methanol, and methanol to respectively obtain saturates, aromatics, and polars [32,33].

Nevertheless, to our knowledge no studies have investigated the repeatability of the SLC as a bio-oil fractionation tool. In addition, in the reported studies, bio-oil is fractionated according to its chemical functions, i.e., esters, aromatics or saturates. In this study, we developed a solid-liquid chromatography protocol, using silica cartridge with flash chromatography device, using eluents of increasing polarity. Bio-oil fractionation by SLC was repeated seven times. Chromatographic and gravimetric data were used to assess method repeatability. Moreover, molecular composition of the collected fractions was characterized by ultra-high-resolution Fourier transform ion cyclotron resonance mass spectrometry (FT-ICR MS) and nuclear magnetic resonance (NMR), as represented by Fig. 1. The molecular and functional differences between the different fractions were evidenced and the SLC repeatability was assessed with fractions obtained under the same conditions.

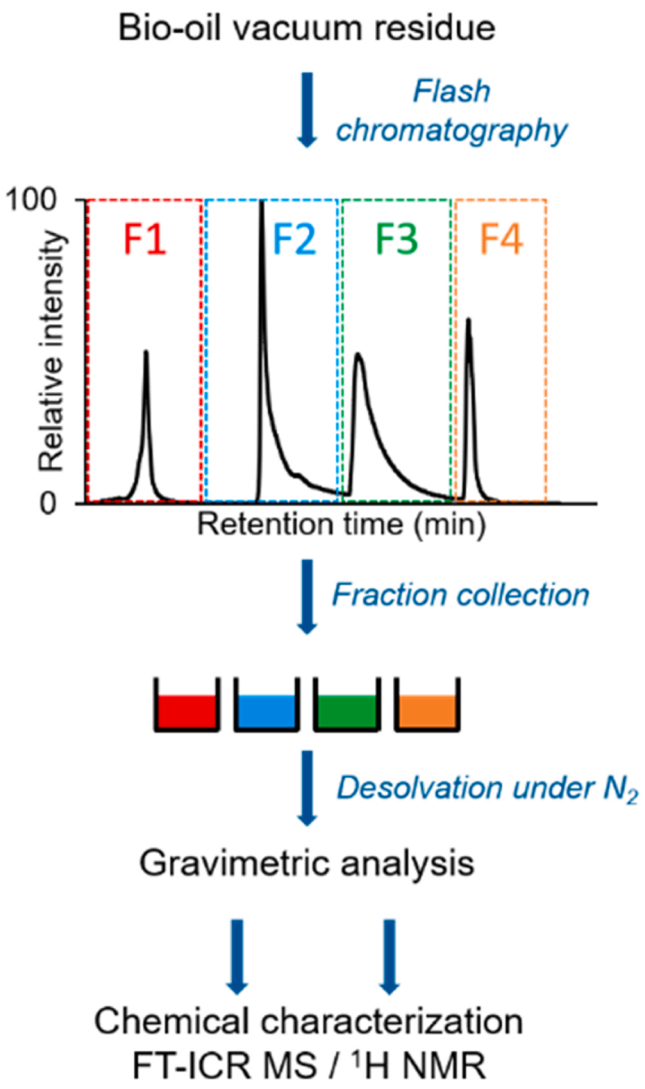


Fig. 1. Scheme of the experimental workflow from sample fractionation to gravimetric analysis and molecular characterization.

2. Materials and methods

2.1. Samples and chemicals

Ultrapure water and HPLC grade solvents were used for this study. A bio-oil from fast pyrolysis of pinewood residue pellets was subject to vacuum distillation using a rotating evaporator at 80 mbar and heated at 70 °C. The resulting bio-oil vacuum residue (VR) was used to demonstrate the SLC procedure and assess its repeatability. This type of sample was chosen for fractionation as it contains lower water and volatile compound amounts compared to regular bio-oil. Indeed, recovery percentage calculated after solvent desolvation will be less biased because of water and light compound evaporation during desolvation step. Thus, water amount in raw bio-oil and resulting vacuum residue was measured by Karl-Fischer titration that yielded 22.72 wt% and 2.82 wt%, respectively. Thermogravimetric analysis enabled to quantify light compound fraction (boiling point < 110 °C) in both samples with 28.3 wt% in the raw bio-oil and 20.1 wt% in the vacuum residue. Elemental composition of the bio-oil and vacuum residue is given in Table S1.

Seven aliquots of VR were prepared and stored in fridge at 4 °C. Before each SLC, 0.5 g of VR was dissolved in 16 g of toluene/methanol mixture (75/25 w/w).

2.2. Bio-oil fractionation by solid-liquid chromatography

Solid-liquid chromatography runs were performed on a Puri-Flash®430 system (Interchim, Montluçon, France) coupled to an evaporative light scattering detector (Sedex 80LT, SEDERE, France). Settings of the instruments are controlled via Interchim software (version 5.0). For each run, 2 mL of the VR solution, corresponding on average to 62 mg of VR, were loaded on silica cartridge (Puriflash, Interchim) to be fractionated (Table 1). Fractionation was processed with 200 mL 100% toluene, 200 mL 90/10 toluene/methanol (v/v), 200 mL 75/25 toluene/methanol (v/v), and 200 mL 90/10/1 methanol/water/formic acid (v/v/v). Eluted bio-oil compounds were automatically collected in test tubes. Solutions in test tubes of same mobile phase composition were merged to form the four different fractions. Liquid part of the fractions was evaporated under nitrogen flow, and desolvation was considered attained when vial weight remained stable.

From the seven SLC performed, three batches, i.e., 12 fractions, were used for FT-ICR MS characterization, while three other batches, for NMR study, the last batch was kept as backup.

2.3. Fourier transform ion cyclotron resonance mass spectrometry

The whole bio-oil vacuum residue and the twelve fractions were investigated by FT-ICR MS with electrospray source (ESI) in negative-ion mode. Measurements were performed on a hybrid quadrupole FT-ICR instrument (Solarix XR, Bruker Daltonics, Bremen, Germany) equipped with a 12 T superconducting magnet and parameters optimized via software FTMS-Control V 2.2.0 (Bruker Daltonics). Prior acquisition, mass spectrometer was externally calibrated with sodium trifluoroacetate solution. The bio-oil VR and solid residues, obtained after fraction evaporation, were dissolved in methanol for a final concentration of 0.3 mg/mL. Sample was infused into the ion source with a 150 μL/h flow rate and the capillary voltage was set at 4.5 kV. The drying gas temperature and flow rate were kept at 200 °C and 4 L/min, respectively, and the nebulizer pressure at 1 bar. Mass spectra were acquired with a 4 megaword time-domain, over a mass range of m/z 92 – 1000, and resulted from 200 scans acquisition.

2.4. FT-ICR MS data treatment

DataAnalysis 5.1 (Bruker Daltonics) software was used to internally calibrate the FT-ICR mass spectra with a list of known bio-oil compounds with a standard deviation of mass error $\leq$ 500 ppb. Peak lists at signal-to-noise ratio ≥ 4 of the different mass spectra were exported and aligned into a matrix according to their m/z value with a 1 ppm tolerance window. For triplicate samples, only values observed at least in 2 samples were kept. The m/z values were assigned with unique molecular formula using PetroOrg software (Florida State University), with C, H, N, O, and S elements, and with $\pm$ 1 ppm tolerance. Deeper sample description was achieved by representing the oxygen atom count distribution of CHO_x molecular series. H/C and O/C van Krevelen diagrams were generated, which enable to distinguish biochemical classes such as lipids, lignin and carbohydrate derivatives [13]. DBE (double bound equivalent) versus carbon number graphs were also made. DBE represents the aromaticity and unsaturation degree of a molecule and is calculated as given by Eq. (1), with n_C, n_H and n_N number of carbon, hydrogen, and nitrogen atoms, respectively.

$$DBE = 1 + n_C - \left(\frac{n_H}{2}\right) + \left(\frac{n_N}{2}\right) \quad (1)$$

Principal component analysis (PCA) of the ESI (−) FT-ICR MS data were performed with Perseus software. It was aimed at assessing similarities in molecular composition description of samples obtained under the same SLC conditions, and therefore to confirm the SLC process repeatability.

Loss of compounds over the fractionation process and molecular specificity of each fraction were investigated by achieving Venn diagrams [34]. Thus, in a first Venn diagram, the FT-ICR MS molecular assignments of the VR bio-oil were compared to those obtained with the four fractions. It ensured to estimate the recovery percentage in terms of molecular assignments before and after the fractionation step. A second Venn diagram was done the same way but without considering the VR bio-oil. Thus, the features exclusively observed in each fraction were plotted on van Krevelen diagrams and DBE versus carbon number maps, as well as the calculation of the weighted averages, in order to highlight their characteristics.

2.5. Nuclear magnetic resonance spectroscopy

As with FT-ICR MS, a total of twelve fractions were characterized by ^{1}H NMR. First, the solid residues of each fraction were dissolved with 500 mg deuterated methanol to be transferred to 5 mm-outer diameter NMR tubes. The ^{1}H NMR spectra were acquired on a Bruker 400 MHz spectrometer (Bruker Avance III) equipped with a 10 mm gradient Broadband Observe (BBO) probe. A 30° pulse - acquisition sequence was used with a 8.6 μs pulse and a relaxation delay of 2.3 s. Number of recorded scans was adapted to the VR amount provided, thus 1400 scans for 8 mg of sample were acquired, on 32 K data points over a 20-ppm spectral width, for 16 K processed data points.

TopSpin 3.6pl5 (Bruker BioSpin, Germany) software was used for spectra analysis. The free induction decay was line-broaded to 0.3 Hz before Fourier transformation, phased to zeroth and first order and chemical shift calibration was done against the residual methyl proton peak of the solvent.

Raw data were extracted to be merged in a matrix for treatment after removal of signals of residual water at 4.7 ppm and methanol solvent at 3.27 ppm. Then, as for FT-ICR MS data, NMR data were subject to PCA with Perseus software, to assess (i) molecular similarities between samples eluted within the same conditions and, (ii) SLC repeatability.

3. Results and discussion

3.1. Repeatability of the fractionation process

For the seven SLC experiments, amount of bio-oil deposited and collected after desolvation were reported in Table 1. The weights of desolvated fractions obtained under the same SLC conditions were averaged and values reported in Table 2**.** The coefficients of variation (CV) obtained for fractions 1, 2, and 3 are equal or below 23% evidencing a very good repeatability of the gravimetric analysis and therefore of the SLC process for these 3 fractions (Table 2). For the fraction 4, the CV calculated from the seven masses is much higher

Table 1
Masses in mg of vacuum residue bio-oil deposited for SLC, total amount of desolvated F1, F2, F3, and F4 fractions collected, and the recovery yield calculated from the ratio of these two values.

	Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Run 7	Mean	SD	CV (%)
Deposited VR sample (mg)	62	56	63	56	61	70	68	**62**	**5**	**9**
Amount of collected fractions* (mg)	50	69	58	47	44	53	55	**54**	**8**	**15**
Recovery yield (%)	81	125	92	84	72	76	81	**87**	**18**	**20**

*fractions F1, F2, F3, and F4.

Table 2
Mean, standard deviation (SD), and coefficient of variation (CV) values calculated for weight of desolvated fractions 1, 2, 3, and 4 (in mg), normalized weight (in %), and normalized fraction areas obtained from chromatogram (in %). Values obtained from the 7 SLC.

	Desolvated fractions (mg)			Normalized weights (%)			Normalized area (%)		
	Mean	SD	CV (%)	Mean	SD	CV (%)	Mean	SD	CV (%)
Fraction 1	6.6	1.5	23	12.4	3.4	27	12.3	1.9	15
Fraction 2	16.4	2.3	14	31.1	7.0	23	34.7	4.5	13
Fraction 3	14.3	1.9	13	26.9	5.0	19	33.0	4.3	13
Fraction 4	16.6	9.2	55	29.6	14.8	50	20.0	9.9	50

(55%). This can be attributed to contaminants absorbed on the silica gel that are eluted or due to elution of some silica particles with the acidic mobile phase [35,36]. Thus, fraction 4 content and therefore weight appeared to be less controlled than what obtained in fractions 1–3, and it lowers repeatability of the process. In fact, when only fractions 1–3 are considered, their normalized weight led to CV $\leq$ 12%, confirming a very good repeatability of the fractionation over the 7 SLC processes (Table S2). Each fraction respectively represented on average 12.4%, 31.1%, 26.9% and 29.6% of the total weight of the desolvated fractions (Table 2). These values were then compared to normalized values of the area underneath the corresponding chromatogram peak and reported in Table 2. Interestingly, both normalized gravimetric and chromatographic values evolve similarly with fractions 1, 2, 3, and 4 corresponding to 12.3%, 34.7%, 33.0% and 20.0% of the chromatogram area, respectively.

This also agrees with the fact that the variability observed for the fraction 4 is due to eluted compounds and not variability in the evaporating process. Thus, it is certainly the elution of more or less silica and/or contaminants from one cartridge to another, with the acidic mobile phase, that is responsible for higher CV.

Recovery yield was calculated for each SLC experiment and resulted to an average of 87%, with a CV of 20% (Table 1). However due to the high variability of the fraction 4, at both the gravimetric and chromatographic levels, it is difficult to estimate its real contribution to the bio-oil composition. Therefore, the recovery yield obtained without fraction 4 was on average 60%, with a CV of 19% (Table S3). Overall, very good repeatability of the SLC process was obtained regarding the amount of collected fractions (CV = 15%) and therefore, the recovery yield (CV = 20%). It is worth noting that the desolvation process is not automatized contrary to the fractionation and collection steps. The observed loss of material from the recovery yield can be explained by their retention over the cartridge, and loss during the different solution transfers and after ELS detection, and to a lesser extent to evaporation of volatile bio-oil compounds. However, the Venn diagram made from the FT-ICR MS molecular assignments of bio-oil VR and its 4 fractions, evidenced that 94% (306/4735) of the features were commonly observed in the five samples (Fig. 2). This suggests a limited loss of the original material, for compounds of molecular weight higher than 92 Da

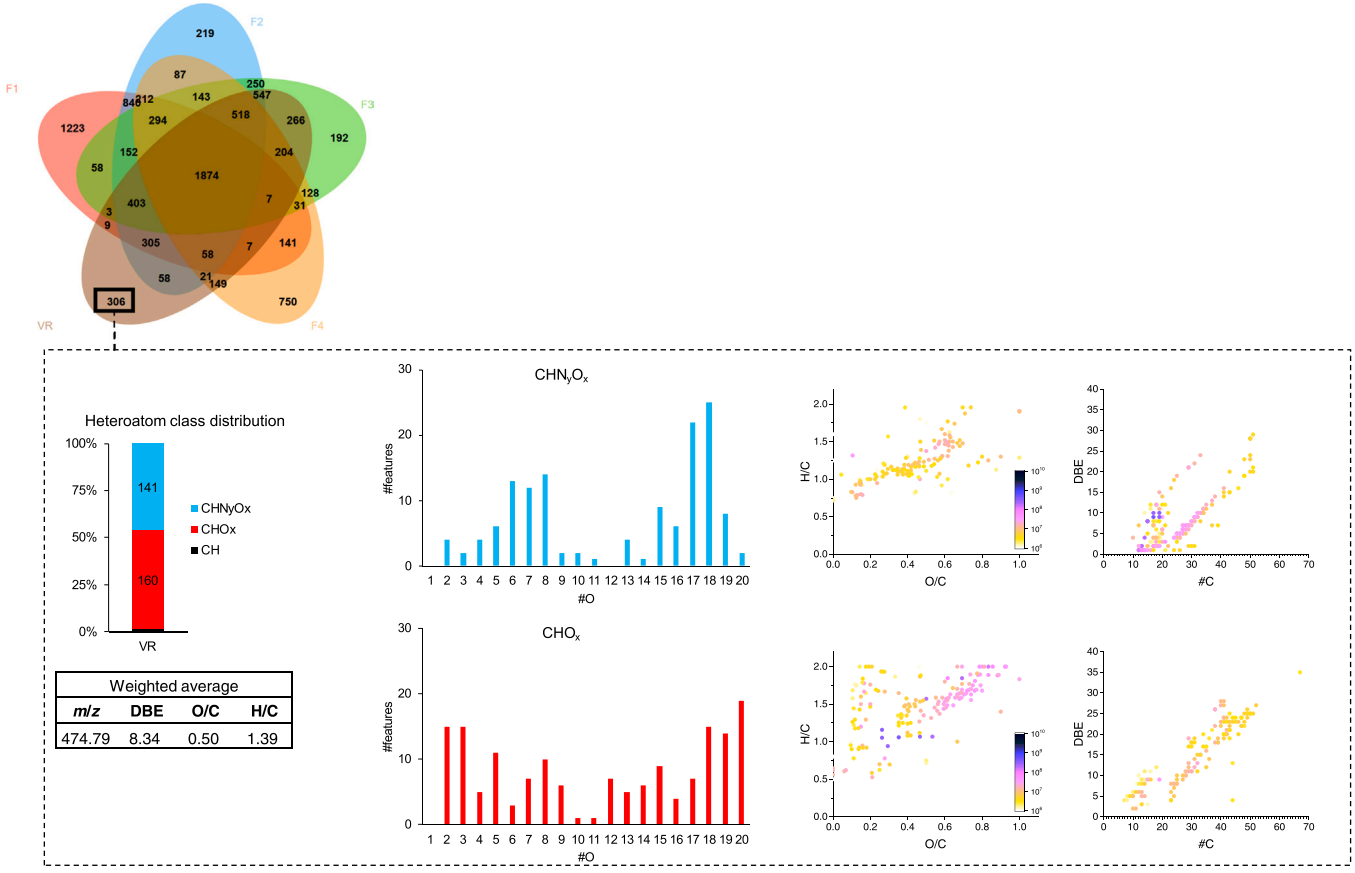


Fig. 2. Venn diagram obtained from the molecular assignments of bio-oil vacuum residue and its fractions 1, 2, 3, and 4 analyzed by ESI (−) FT-ICR MS. In the inset, the features specific to bio-oil VR are represented according to their heteroatom class distribution; the weighted averages calculated for the *m/z*, DBE, O/C and H/C; oxygen atom count distribution for CHO_x and CHN_yO_x molecular series, and their corresponding van Krevelen diagram and DCBE vs. #C.

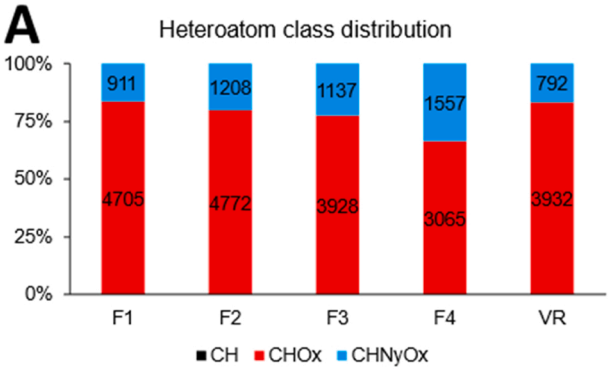
(value corresponding to the low-mass cutoff of the FT-ICR mass spectrometer). The 306 compounds specifically observed in the bio-oil VR were plotted on different graphs (Fig. 2). However, these graphs did not evidence any trend, such as low-mass species, which would agree with the evaporation of the low-mass compounds in the fractions (Fig. 2). But some of these species comprise middle to high number oxygen atoms, which might be responsible for irreversible retention over silica.

Eventually, results obtained from both gravimetric and chromatographic analyses demonstrated that weight contribution of each bio-oil fraction can be obtained from direct integration of chromatographic peak area. This would avoid long tedious desolvation step for gravimetric analysis, which is time and gas consuming, and would avoid material loss. Besides, chromatographic information can also be used to monitor efficiency of upgrading treatments. Indeed, such treatments are often aimed at reducing oxygen amount and increasing higher heating values (HHV) of the bio-oil [37]. Thus, one may expect to have a stronger contribution of the fraction 1 and a smaller one from fractions 2–4, to the bio-oil composition after such treatment.

3.2. Molecular characterization by FT-ICR MS

Assignment of the FT-ICR *m/z* signals of the studied samples led to CH, CHO_x, and CHN_yO_x molecular series (Fig. 3A). The heteroatom class distributions were quite similar for all studied samples with 66–84% of the contributions relative to CHO_x molecular series, and CHN_yO_x features corresponding to 16–34% of the assignments. CHN_yO_x class was more represented in fraction 4, compared to the other samples, which can be attributed to formic acid from mobile phase, that was showed to favor detection of these species even in negative-ion mode [38].

Some molecular descriptors were also investigated by calculating the weighted averages of *m/z*, O/C, H/C, and DBE values from all molecular assignments of the different samples (Fig. 3B). Thus, an increase of the mobile phase polarity led to species of higher H/C and O/C values, and lower DBE. Fig. 4 confirms the previous observations. The CHO_x distribution according to their oxygen atom count extends and is centered



B

	Weighted average			
	m/z	**DBE**	**O/C**	**H/C**
F1	436.00	11.95	0.27	1.14
F2	448.74	10.66	0.37	1.21
F3	449.46	9.09	0.50	1.31
F4	391.75	7.31	0.61	1.36
VR	431.14	8.18	0.50	1.34

Fig. 3. (A) Heteroatom class distribution (values are number of assignments) and (B) weighted averages of *m/z*, DBE, O/C, and H/C, calculated from all assignments obtained by FT-ICR MS analysis of the four fractions and bio-oil VR.

at higher values in the most polar fractions (3 and 4). Van Krevelen diagrams demonstrates that in fraction 1, the most intensely detected species are low-oxygenated unsaturated hydrocarbons, from lignin pyrolysis, with DBE ~ 10. For fraction 2, the most intensely detected compounds are slightly shifted to higher O/C and lower DBE values, compared to fraction 1, but these properties are still related to pyrolytic lignin. For fractions 3 and 4, the shift to higher O/C and lower DBE values is even more pronounced, with the most abundant species related to pyrolytic cellulose and hemicellulose.

Polarity of the mobile phase influences the molecular fingerprint of the fractions. This is also observed in the PCA score plots done with the ESI (−) FT-ICR MS data (Fig. 5). Within the first component, samples are ordered according to the polarity parameter. Moreover, the PCA also shows that fractions obtained under the same elution conditions are clustering together confirming the process repeatability at the molecular level.

A second Venn diagram (Fig. S1) was generated from the assignments obtained for the four fractions in order to highlight fraction specificities. Features exclusively observed in each fraction were extracted and represented on various graphs. These compounds are mainly CHO_x species, in fractions 1, 2, and 3, while in fraction 4, they are CHN_yO_x components, which was previously be explained by the acidic medium of the mobile phase [38] (Fig. S2A). Molecular descriptors (Fig. S2B), van Krevelen diagrams, and DBE vs. carbon number maps (Fig. 6) emphasize what was obtained above with all features. Briefly, DBE values decrease and O/C and H/C values increase with phase polarity increase. Thus, fraction 1 is characterized by exclusive highly unsaturated and poorly oxygenated compounds. In fraction 2, the components are less unsaturated and more oxygenated compared to fraction 1. More oxygenated species, carbohydrates from cellulose and hemicellulose pyrolysis, are specifically observed in fraction 3. This is even more marked in fraction 4 where detected species are more polar.

Molecular characterization by NMR.

Proton NMR analysis allowed investigating the functional groups of the compounds comprising the four different fractions. Comparison of the achieved NMR spectra (Fig. 7) allowed highlighting some differences between the fractions. Fractions 1 and 2 present similar functional groups typical of lignin compounds with aldehydes, ethers, methoxy, alcohols, aliphatics, and alkanes [39,40]. Aromatic region (6–8.5 ppm; area B on the Fig. 7) is richer in the fraction 1 than in the fraction 2. This latter is characterized by more peaks in the area C of the Fig. 7 corresponding to oxygenated functions. These observations agree with results obtained by FT-ICR MS, where fractions 1 and 2 are characterized by lignin derivatives at different oxygenation degree. For fractions 3, signals of $\delta > 6$ ppm are poorly observed, indicating that chemical functions related to poorly oxygenated lignin derivatives are not prominent. However, intense signals are observed in the region extending from 3 to 6 ppm that can be attributed to alcohols related to carbohydrates but also to more polar lignin compounds comprising also ether and methoxy groups [39–41]. This goes in line with the previous statement made from the van Krevelen diagram of the fractions 3. Finally, regarding the fractions 4, the low amount of material was responsible for the little information available on the spectra. The signal around 8.3 ppm can be attributed to formic acid from mobile phase used for fraction 4 extraction [42]. Eventually, NMR spectra obtained for same fractions are quite similar, which demonstrates good repeatability of the SLC process This was also confirmed by the PCA score plots (Fig. S3) where the fractions obtained under the same conditions clustered together and were also ordered, within the first component, according to their polarity.

3.3. Assets of fractionation for bio-oil characterization and upgrading

Separative process combined with direct injection mass spectrometry is employed was already reported to reduce molecular complexity of complex matrices, and therefore, to allow a better insight into the composition of these samples [13]. If most of the molecular assignments

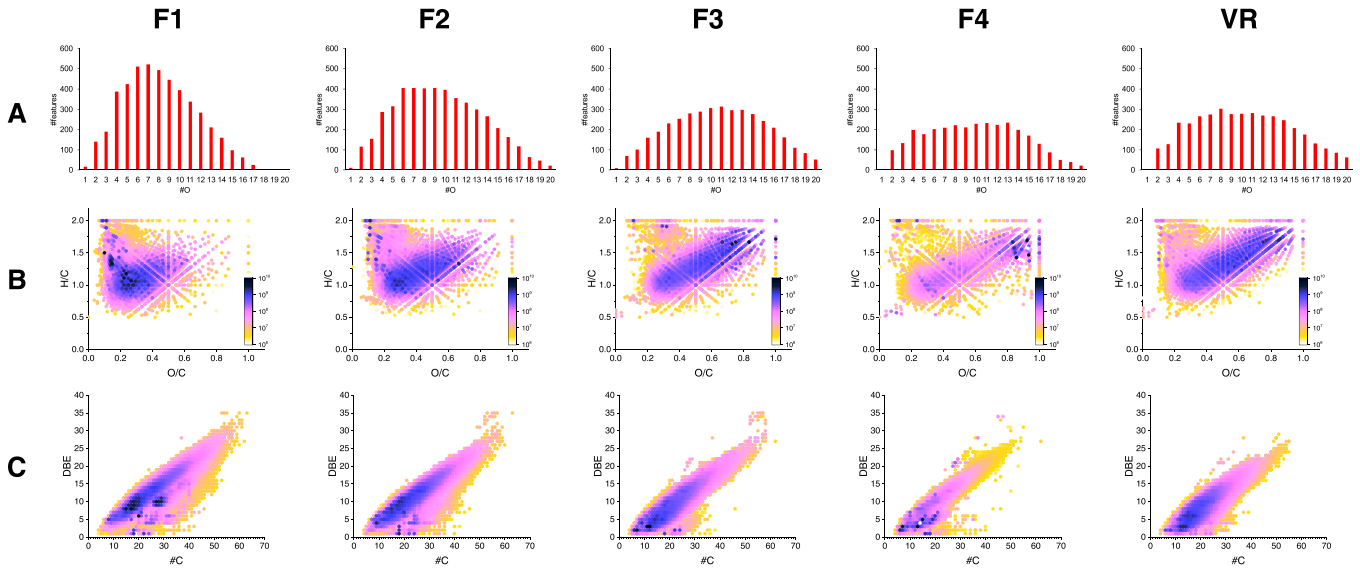


Fig. 4. (A) Oxygen atom count distributions, (B) van Krevelen diagrams, and (C) DBE vs. #C maps of the CHO_x formulae achieved for F1, F2, F3, F4, and VR samples analyzed by ESI (−) FT-ICR MS.

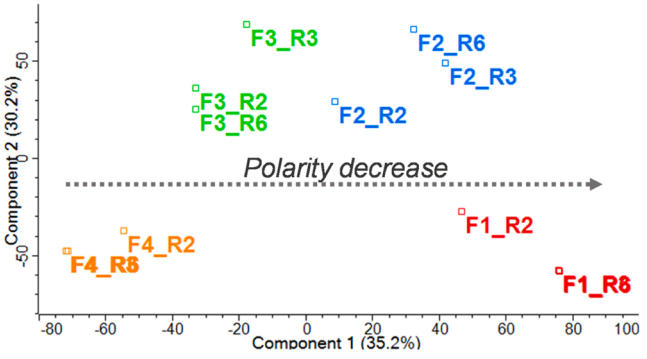


Fig. 5. Principal component analysis (PCA) score plot of all data obtained from the fractions analyzed by ESI (−) FT-ICR MS.

obtained by ESI (−) FT-ICR MS of bio-oil VR is observed within the four fractions, each fraction revels additional components (Fig. 2). This can be explained by reduction of ion suppression phenomenon thanks to the upstream separation step, which favor detection of minor components as showed in Fig. S4. Thus, flash chromatography appears to be an efficient tool to have a more detailed molecular composition description of complex organic mixtures such as bio-oil. Besides, it was demonstrated that it was possible to extract some quantitative information from the acquired chromatograms, which information is missing in direct infusion FT-ICR MS.

Concerning the upgrading prospects, fractions obtained as part of this study possess molecular particularities that can be valorized in various ways. Thus, compounds of fraction 1 and, to a lesser extent, of fraction 2 can be potentially upgraded into biofuels [43,44]. However, fraction 2 was showed to comprise oxygenated aromatics such as polyphenolics, and therefore could be a source of antioxidants [45,46]. For instance, vanillin, whose formula is $C_8H_8O_3$, was putatively found in

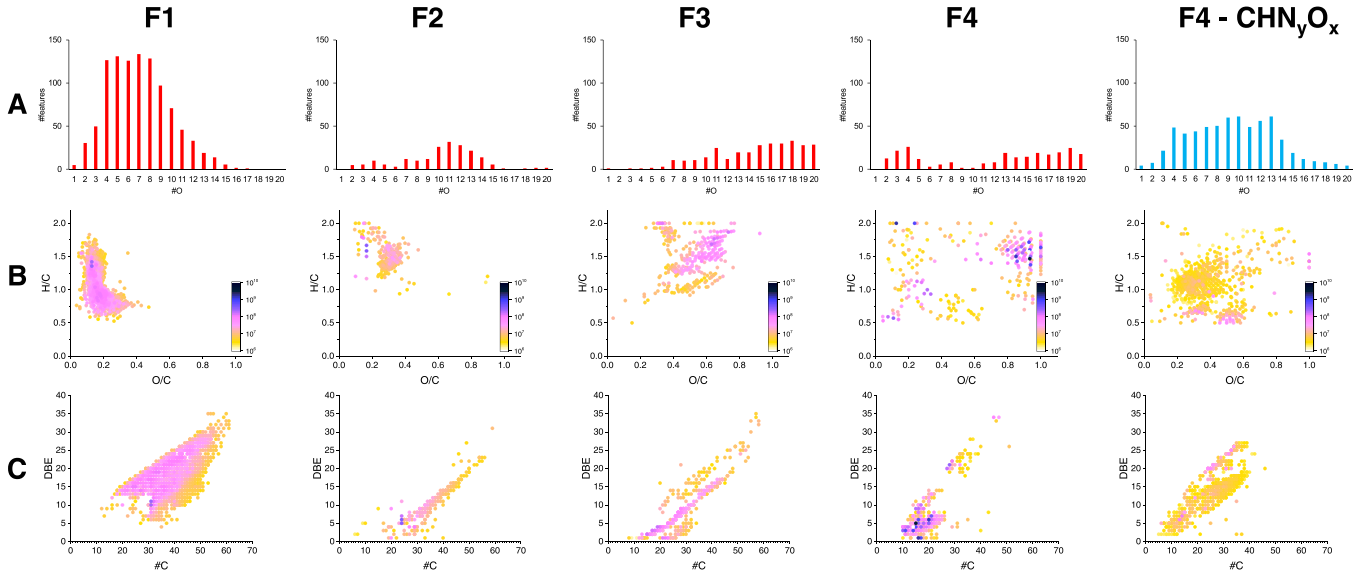
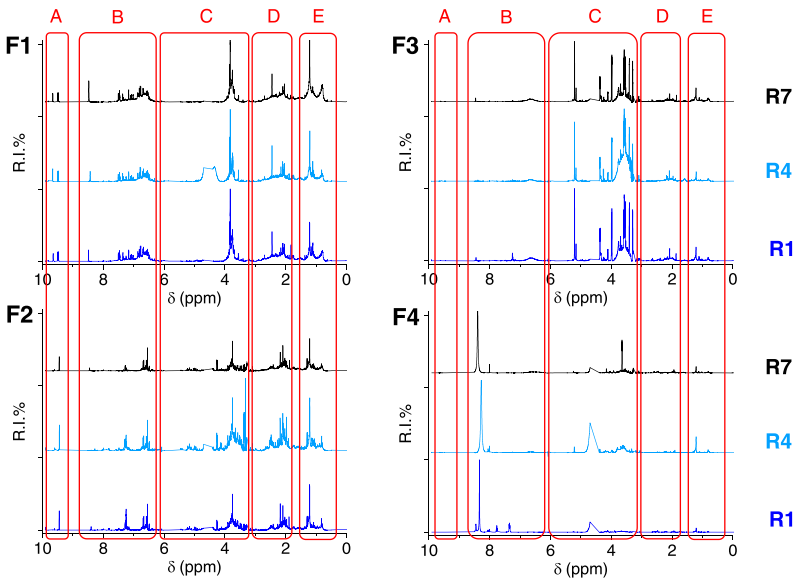


Fig. 6. (A) Oxygen atom count distributions, (B) van Krevelen diagrams, and (C) DBE vs. #C maps of the CHO_x (in red) and CHN_yO_x (in blue - for fraction 4) molecular formulae exclusively observed in fractions 1, 2, 3, and 4 after ESI (−) FT-ICR MS analysis.



Group	Functional groups
A	Aldehydes
B	Aromatics and heteroaromatics
C	Ethers, methoxy, alcohols
D	Aliphatic α-to unsaturation or heteroatom
E	Alkanes

Fig. 7. Proton-NMR spectra achieved for the four fractions, measured in triplicate, after removal of the signals attributed to water at 4.6 ppm and to methanol at 3.27 ppm, with regions of interest according to functional groups. * formic acid signal.

fractions 1 and 2. Putative dehydrated carbohydrates, levoglucosan and cellobiosan, were found in fraction 3 and fraction 4 at high intensity. These compounds can be upgraded into value-added chemicals such as ethanol and organic acids, and by extension, into biofuels [3]. Notably, levoglucosan raises awareness due its high amount in bio-oils and because of its potential applications as pesticides, antibiotics, and bio-ethanol [26,47].

4. Conclusion

In this study, flash chromatography was applied to separate bio-oil vacuum residue into four distinct fractions. Silica cartridge was used, which presents some advantages regarding the cost and the use. Good repeatability of this fractionation process was established at both the gravimetric and chromatographic levels, especially on fractions 1, 2, and 3. Fraction 4 presented more variability that was explained by elution of contaminants or silica from the cartridge by the acidic mobile phase. Good repeatability of the flash chromatography process was also demonstrated at the molecular level, after characterization of the fractions by FT-ICR MS and NMR. These two techniques also evidenced the molecular specificities of each of the bio-oil fractions. Fraction 1 was characterized by low-oxygenated and unsaturated compounds, fraction 2 with more oxygenated unsaturated compounds, fraction 3 comprises some cellulose and hemicellulose derivatives, and fraction 4 with more polar compounds from cellulose and hemicellulose pyrolysis. Thus, flash chromatography fractionation was showed to present several advantages on the analytical point of view by offering a better insight into the molecular composition of bio-oil, and to provide fractions that can be upgraded into green fuels or value-added chemicals.

CRediT authorship contribution statement

Jasmine Hertzog: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Carole Garnier:** Conceptualization, Methodology, Validation, Investigation, Writing – review & editing, Visualization. **Charlotte Mase:** Conceptualization, Investigation, Writing – review & editing, Visualization. **Sandra Mariette:** Investigation, Writing – review & editing, Visualization. **Olivier Serve:** Investigation, Writing – review & editing, Visualization. **Marie Hubert-Roux:** Investigation, Writing – review & editing. **Carlos Afonso:** Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Pierre Giusti:** Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Caroline Barrère-Mangote:** Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work has been partially supported by University of Rouen Normandy, INSA Rouen Normandy, the Centre National de la Recherche Scientifique (CNRS), European Regional Development Fund (ERDF), Labex SynOrg (Grant ANR-11-LABX-0029), Carnot Institut I2C, the Graduate School for Research XL-Chem (Grant ANR-18-EURE-0020 XL CHEM), and by Région Normandie.

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jaap.2022.105611.

References

[1] A. Demirbaş, Mechanisms of liquefaction and pyrolysis reactions of biomass, Energy Convers. Manag. 41 (2000) 633–646, https://doi.org/10.1016/S0196-8904(99)00130-2.
[2] F.J. Sánchez-Borrego, P. Álvarez-Mateos, J.F. García-Martín, Biodiesel and other value-added products from bio-oil obtained from agrifood waste, Processes 9 (2021) 797, https://doi.org/10.3390/pr9050797.
[3] A.P. Pinheiro Pires, J. Arauzo, I. Fonts, M.E. Domine, A. Fernández Arroyo, M. E. Garcia-Perez, J. Montoya, F. Chejne, P. Pfromm, M. Garcia-Perez, Challenges and opportunities for bio-oil refining: a review, Energy Fuels 33 (2019) 4683–4720, https://doi.org/10.1021/acs.energyfuels.9b00039.
[4] D.C. Elliott, G.F. Schiefelbein, Liquid hydrocarbon fuels from biomass, American Chemical Society, Division of Fuel Chemistry Annual Meeting Preprints, 1989, pp. 1160–1166.
[5] J. Meng, A. Moore, D.C. Tilotta, S.S. Kelley, S. Adhikari, S. Park, Thermal and storage stability of bio-oil from pyrolysis of torrefied wood, Energy Fuels 29 (2015) 5117–5126, https://doi.org/10.1021/acs.energyfuels.5b00929.

[6] W. Cai, N. Kang, M.K. Jang, C. Sun, R. Liu, Z. Luo, Long term storage stability of bio-oil from rice husk fast pyrolysis, Energy 186 (2019), 115882, https://doi.org/10.1016/j.energy.2019.115882.

[7] J. Hertzog, V. Carré, F. Aubriet, Chapter 22 – Contribution of Fourier transform mass spectrometry to bio-oil study, in: B. Kanawati, P. Schmitt-Kopplin (Eds.), Fundamentals and Applications of Fourier Transform Mass Spectrometry, Elsevier, 2019, pp. 679–733, https://doi.org/10.1016/B978-0-12-814013-0.00022-3.

[8] I. Graca, J.M. Lopes, H.S. Cerqueira, M.F. Ribeiro, Bio-oils upgrading for second generation biofuels, Ind. Eng. Chem. Res. 52 (2013) 275–287, https://doi.org/10.1021/ie301714x.

[9] M.I. Jahirul, J.R. Brown, W. Senadeera, N. Ashwath, C. Laing, J. Leski-Taylor, M. G. Rasul, Optimisation of bio-oil extraction process from beauty leaf (Calophyllum inophyllum) oil seed as a second generation biodiesel source, Procedia Eng. 56 (2013) 619–624, https://doi.org/10.1016/j.proeng.2013.03.168.

[10] M. Stas, D. Kubicka, J. Chudoba, M. Pospisil, Overview of analytical methods used for chemical characterization of pyrolysis bio-oil, Energy Fuels 28 (2014) 385–402, https://doi.org/10.1021/ef402047y.

[11] C.M. Michailof, K.G. Kalogiannis, T. Sfetsas, D.T. Patiaka, A.A. Lappas, Advanced analytical techniques for bio-oil characterization, Wiley Interdiscip. Rev. Energy Environ. 5 (2016) 614–639, https://doi.org/10.1002/wene.208.

[12] Y. Wang, Y. Han, W. Hu, D. Fu, G. Wang, Analytical strategies for chemical characterization of bio-oil, J. Sep. Sci. 43 (2020) 360–371, https://doi.org/10.1002/jssc.201901014.

[13] J. Hertzog, C. Mase, M. Hubert-Roux, C. Afonso, P. Giusti, C. Barrère-Mangote, Characterization of heavy products from lignocellulosic biomass pyrolysis by chromatography and fourier transform mass spectrometry: a review, Energy Fuels (2021), https://doi.org/10.1021/acs.energyfuels.1c02098.

[14] J.L. Maître, M. Hubert-Roux, B. Paupy, S. Marceau, C.P. Rüger, C. Afonso, P. Giusti, Structural analysis of heavy oil fractions after hydrodenitrogenation by high-resolution tandem mass spectrometry and ion mobility spectrometry, Faraday Discuss. 218 (2019) 417–430, https://doi.org/10.1039/C8FD00239H.

[15] C.P. Rüger, J. Maillard, J. Le Maître, M. Ridgeway, C.J. Thompson, I. Schmitz-Afonso, T. Gautier, N. Carrasco, M.A. Park, P. Giusti, C. Afonso, Structural study of analogues of Titan's haze by trapped ion mobility coupled with a Fourier transform ion cyclotron mass spectrometer, J. Am. Soc. Mass Spectrom. 30 (2019) 1169–1173, https://doi.org/10.1007/s13361-019-02205-7.

[16] C. Reymond, A. Dubuis, A. Le Masle, C. Colas, L. Chahen, E. Destandau, N. Charon, Characterization of liquid–liquid extraction fractions from lignocellulosic biomass by high performance liquid chromatography hyphenated to tandem high-resolution mass spectrometry, J. Chromatogr. A 2020 (1610), 460569, https://doi.org/10.1016/j.chroma.2019.460569.

[17] C.P. Rüger, J. Le Maître, J. Maillard, E. Riches, M. Palmer, C. Afonso, P. Giusti, Exploring complex mixtures by cyclic ion mobility high-resolution mass spectrometry: application toward petroleum, Anal. Chem. 93 (2021) 5872–5881, https://doi.org/10.1021/acs.analchem.1c00222.

[18] C. Reymond, A. Le Masle, C. Colas, N. Charon, A rational strategy based on experimental designs to optimize parameters of a liquid chromatography-mass spectrometry analysis of complex matrices, Talanta 205 (2019), 120063, https://doi.org/10.1016/j.talanta.2019.06.063.

[19] M. Sarrut, A. Corgier, G. Crétier, A. Le Masle, S. Dubant, S. Heinisch, Potential and limitations of on-line comprehensive reversed phase liquid chromatography×supercritical fluid chromatography for the separation of neutral compounds: an approach to separate an aqueous extract of bio-oil, J. Chromatogr. A 1402 (2015) 124–133, https://doi.org/10.1016/j.chroma.2015.05.005.

[20] A.A. Kiss, J.-P. Lange, B. Schuur, D.W.F. Brilman, A.G.J. van der Ham, S.R. A. Kersten, Separation technology–making a difference in biorefineries, Biomass Bioenergy 95 (2016) 296–309, https://doi.org/10.1016/j.biombioe.2016.05.021.

[21] X. Hu, D. Mourant, R. Gunawan, L. Wu, Y. Wang, C. Lievens, C.-Z. Li, Production of value-added chemicals from bio-oil via acid catalysis coupled with liquid-liquid extraction, RSC Adv. 2 (2012) 9366–9370, https://doi.org/10.1039/c2ra21597g.

[22] H.-J. Huang, S. Ramaswamy, Separation and purification of phytochemicals as co-products in biorefineries, Biorefinery Co-Products, John Wiley & Sons, Ltd, n.d., pp. 37–53. ⟨https://doi.org/10.1002/9780470976692.ch3⟩.

[23] H.L. Chum, R.E. Kreibich, Process for Preparing Phenolic Formaldehyde Resole Resin Products Derived from Fractionated Fast-pyrolysis Oils. US Patent 5. US 5091499 A, US 5091499 A, 1992. ⟨https://www.google.com/patents/US5091499⟩.

[24] G. Underwood, R.G. Graham, Method of Using Fast Pyrolysis Liquids as Liquid Smoke, US Patent 4. US 4876108 A., US 4876108 A, 1989. ⟨https://www.google.ch/patents/US4876108⟩.

[25] C. Lindfors, E. Kuoppala, A. Oasmaa, Y. Solantausta, V. Arpiainen, Fractionation of bio-oil, Energy Fuels 28 (2014) 5785–5791, https://doi.org/10.1021/ef500754d.

[26] Y.H. Chan, S.K. Loh, B.L.F. Chin, C.L. Yiin, B.S. How, K.W. Cheah, M.K. Wong, A.C. M. Loy, Y.L. Gwee, S.L.Y. Lo, S. Yusup, S.S. Lam, Fractionation and extraction of bio-oil for production of greener fuel and value-added chemicals: Recent advances and future prospects, Chem. Eng. J. 397 (2020), 125406, https://doi.org/10.1016/j.cej.2020.125406.

[27] J.P. Stanford, P.H. Hall, M.R. Rover, R.G. Smith, R.C. Brown, Separation of sugars and phenolics from the heavy fraction of bio-oil using polymeric resin adsorbents, Sep. Purif. Technol. 194 (2018) 170–180, https://doi.org/10.1016/j.seppur.2017.11.040.

[28] A.E. Putun, A. Ozcan, E. Putun, Pyrolysis of hazelnut shells in a fixed-bed tubular reactor: yields and structural analysis of bio-oil, J. Anal. Appl. Pyrolysis 52 (1999) 33–49, https://doi.org/10.1016/S0165-2370(99)00044-3.

[29] N. Ozbay, B.B. Uzun, E.A. Varol, A.E. Putun, Production and characterization of bio-oil and biochar from rapeseed cake, Fuel Process. Technol. 87 (2006) 1013–1019, https://doi.org/10.1016/j.fuproc.2006.07.009.

[30] D. Ozcimen, F. Karaosmanoglu, Production and characterization of bio-oil and biochar from rapeseed cake, Renew. Energy 29 (2004) 779–787, https://doi.org/10.1016/j.renene.2003.09.006.

[31] P. Das, T. Sreelatha, A. Ganesh, Bio oil from pyrolysis of cashew nut shell-characterisation and related properties, Biomass Bioenergy 27 (2004) 265–275, https://doi.org/10.1016/j.biombioe.2003.12.001.

[32] R.L. Ware, S.M. Rowland, R.P. Rodgers, A.G. Marshall, Advanced chemical characterization of pyrolysis oils from landfill waste, recycled plastics, and forestry residue, Energy Fuels 31 (2017) 8210–8216, https://doi.org/10.1021/acs.energyfuels.7b00865.

[33] R.L. Ware, R.P. Rodgers, A.G. Marshall, O.D. Mante, D.C. Dayton, S. Verdier, J. Gabrielsen, S.M. Rowland, Detailed chemical composition of an oak biocrude and its hydrotreated product determined by positive atmospheric pressure photoionization Fourier transform ion cyclotron resonance mass spectrometry, Sustain. Energy Fuels (2020), https://doi.org/10.1039/C9SE00837C.

[34] P. Bardou, J. Mariette, F. Escudié, C. Djemiel, C. Klopp, jvenn: an interactive Venn diagram viewer, BMC Bioinform. 15 (2014) 293, https://doi.org/10.1186/1471-2105-15-293.

[35] H. Spitz, Impurities in silica gel as a critical factor in the qualitative evaluation of an organic compound, J. Chromatogr. A 42 (1969) 384–388, https://doi.org/10.1016/S0021-9673(01)80638-5.

[36] M. Khalifa, M. Hajji, H. Ezzaouia, Impurity removal process for high-purity silica production by acid leaching, EPJ Web Conf. 29 (2012) 00014, https://doi.org/10.1051/epjconf/20122900014.

[37] C. Muangsuwan, W. Kriprasertkul, S. Ratchahat, C.-G. Liu, P. Posoknistakul, N. Laosiripojana, C. Sakdaronnarong, Upgrading of light bio-oil from solvothermolysis liquefaction of an oil palm empty fruit bunch in glycerol by catalytic hydrodeoxygenation using NiMo/Al2O3 or CoMo/Al2O3 catalysts, ACS Omega 6 (2021) 2999–3016, https://doi.org/10.1021/acsomega.0c05387.

[38] J. Hertzog, V. Carré, Y. Le Brech, A. Dufour, F. Aubriet, Toward controlled ionization conditions for ESI-FT-ICR-MS analysis of bio-oils from lignocellulosic material, Energy Fuels 30 (2016) 5729–5739, https://doi.org/10.1021/acs.energyfuels.6b00655.

[39] J. Joseph, C. Baker, S. Mukkamala, S.H. Beis, M.C. Wheeler, W.J. DeSisto, B. L. Jensen, B.G. Frederick, Chemical shifts and lifetimes for nuclear magnetic resonance (NMR) analysis of biofuels, Energy Fuels 24 (2010) 5153–5162, https://doi.org/10.1021/ef100504d.

[40] C.A. Mullen, G.D. Strahan, A.A. Boateng, Characterization of various fast-pyrolysis bio-oils by NMR spectroscopy, Energy Fuels 23 (2009) 2707–2718, https://doi.org/10.1021/ef801048b.

[41] Chapter 2 – Structure and characteristics of lignin, in: J. Huang, S. Fu, L. Gan (Eds.), Lignin Chemistry and Applications, Elsevier, 2019, pp. 25–50, https://doi.org/10.1016/B978-0-12-813941-7.00002-3.

[42] H.V. Halleraker, T. Barth, Quantitative NMR analysis of the aqueous phase from hydrothermal liquefaction of lignin, J. Anal. Appl. Pyrolysis 151 (2020), 104919, https://doi.org/10.1016/j.jaap.2020.104919.

[43] R. Shu, R. Li, B. Lin, C. Wang, Z. Cheng, Y. Chen, A review on the catalytic hydrodeoxygenation of lignin-derived phenolic compounds and the conversion of raw lignin to hydrocarbon liquid fuels, Biomass Bioenergy 132 (2020), 105432, https://doi.org/10.1016/j.biombioe.2019.105432.

[44] P. Bi, J. Wang, Y. Zhang, P. Jiang, X. Wu, J. Liu, H. Xue, T. Wang, Q. Li, From lignin to cycloparaffins and aromatics: directional synthesis of jet and diesel fuel range biofuels using biomass, Bioresour. Technol. 183 (2015) 10–17, https://doi.org/10.1016/j.biortech.2015.02.023.

[45] A. Tai, T. Sawano, F. Yazama, H. Ito, Evaluation of antioxidant activity of vanillin by using multiple antioxidant assays, Biochim. Biophys. Acta 2011 (1810) 170–177, https://doi.org/10.1016/j.bbagen.2010.11.004.

[46] F. Stankovikj, A.G. McDonald, G.L. Helms, M.V. Olarte, M. Garcia-Perez, Characterization of the water-soluble fraction of woody biomass pyrolysis oils, Energy Fuels 31 (2017) 1650–1664, https://doi.org/10.1021/acs.energyfuels.6b02950.

[47] Y. Yu, Y.W. Chua, H. Wu, Characterization of pyrolytic sugars in bio-oil produced from biomass fast pyrolysis, Energy Fuels 30 (2016) 4145–4149, https://doi.org/10.1021/acs.energyfuels.6b00464.